Fiscal Year:	FY 2017	**Task Last Updated:**	FY 01/22/2018
PI Name:	Miller, Fletcher J Ph.D.		
Project Title:	Developing the Narrow Channel Apparatus as a NASA Standard Material Flammability Test (NNX14AC70A)		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Combustion science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	92182-1323	**Congressional District:**	53
Comments:	Previously at National Center for Space Exploration Research, NASA Glenn Research Center, until 2007.		
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2009 Combustion Science NNH09ZTT001N
Start Date:	02/01/2014	**End Date:**	01/31/2017
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	3
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
Contact Monitor:	Olson, Sandra	**Contact Phone:**	216-433-2859
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 1/31/2017 (original end date was 1/31/2016) per NSSC information (Ed., 3/14/16)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Olson, Sandra Ph.D. (NASA Glenn Research Center) Wichman, Indrek Ph.D. (Michigan State University)		
Grant/Contract No.:	NNX14AC70A		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NOTE (Ed., January 2014): Continuation of "Developing the Narrow Channel Apparatus as a NASA Standard Material Flammability Test," grant # NNX10AD96A with the same Principal Investigator. This research will study flame spread over flammable solid materials such as plastic, paper, cloth, and hair, using a new apparatus designed to simulate microgravity flow conditions on earth. We designed and built two new Narrow Channel Apparatuses (NCA) at San Diego State University (SDSU), and one at Michigan State (MSU), that control environmental variables such as flow velocity, gas pressure, and oxygen concentration to reproduce space craft atmospheres. Additionally, the Narrow Channel suppresses buoyant flow by having a closely spaced ceiling on the channel, so that flow conditions closely resemble those found on space craft. The objective of the research is to prove the Narrow Channel Apparatus as a viable test method for opposed flow material flammability testing. This will allow NASA to screen materials in more realistic conditions than does the current NASA test which makes no account of microgravity. The experimental research consists of test wherein we ignite various solids in the channels and measure how fast the flame spreads in various conditions. To accompany the experiments, we are developing a numerical model so that we can determine certain quantities such as heat loss to the channel ceiling that are difficult to measure.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our research encompasses flame spread and material flammability in narrow channels or gaps. We use this geometry to simulate microgravity conditions. However, many fire situations involve flames in narrow gaps. TWA Flight 800, which crashed off the coast of Canada, was brought down by a small fire spreading through some narrow gaps in the walls and ceiling near the cockpit, for instance. Many building fires involve fire spread inside walls. Fires in computers likewise can occur between the printed circuit boards. Thus, information gained form experiments about how flames spread in various atmospheric conditions in narrow gaps is useful in areas well beyond spacecraft fire safety. The numerical model we are developing will be validated against our experiments and can also be used to study flame spread in these gaps.
Task Progress:	[Ed. note: compiled from PI's final progress report submitted August 2017] The SDSU NCA has been shown to successfully simulate microgravity for not only thin cellulose fuels, but also for thin Poly(methyl methacrylate) (PMMA) sheets at 1 atm pressure if the proper gap height is chosen. Tests completed in the SDSU NCA closely compare to actual microgravity conditions seen in drop tower experiments under otherwise the same conditions. Three oxygen mole fractions were tested where the NCA corresponded with true microgravity testing. Flame spread rate was shown to be extremely sensitive to oxygen concentration as experiments run at 50% O2 were many times faster than samples burned at 21% O2. These results provide further verification that the SDSU NCA is a viable option to complement or replace NASA's Test 1 for material flammability testing. While gap height experiments have shown 18 mm and 6 mm gaps to be unfit for the SDSU NCA, it is unclear what gap height produces the best compromise in buoyancy suppression and heat loss to the top and bottom plates of the NCA. Sidebotham suggests a full gap height of 10-12 mm for Whatman 44 filter paper using side view image analysis and quantitative methods. Future planned experiments will determine the best gap height for thermally thin PMMA in the NCA. The SDSU NCA has been shown to match the thick PMMA data of the Michigan State University Narrow Channel Apparatus very well at an oxygen concentration of 21 percent with the minor differences being attributed to the false bottom material and the PMMA thickness. Only one true microgravity test was available for comparison. The flame spread velocity found using the SDSU Narrow Channel Apparatus fell close to the true microgravity data point near the lower flame extinction limit. Larger differences in flame spread velocity were found at the extremes of the range tested (11 cm/s and 25 cm/s). At low opposed flow velocities the flame struggles to move forward and often shrinks away from the side walls making it difficult to get accurate measurements. Minor flaws in flow (caused by lifting and warping of the fuel bed) likely affect the flame spread rate more as gas velocity increases creating less precise results at higher speeds. Potential causes for scatter include performing a test while the channel is still warm from the previous test, effects from the aluminum tape used to hold down the sample, effects from the fuel lifting, or effects from the igniter wire being in different areas of the flow. At 21 percent oxygen concentration the flame spread rate rises, reaches a peak, and then begins to once again drop off. This is expected as the flame experiences the three different opposed flow regimes. First, speeding up through the oxygen transport limited regime, followed by a more steady gas phase conduction limited region, and finally slowing in the residence time limited regime on its way to flame blow-off. This shows there is an optimal opposed velocity that maximizes the flame spread rate, or is the most hazardous fire condition. At higher oxygen concentrations the peak flame spread velocity is thought to be out of the range of the tested opposed flow velocity. Oxygen concentration has the largest effect on flame spread rate; however, it has little effect on flame length. Because of this a higher oxygen concentration means a lower residence time. As the residence time increases, the depth of pyrolysis increases and less soot is created, resulting in a bluer flame. In exploring the effect of 2-D conduction we found Fire Dynamics Simulator coupled with Gpyro does model the combustion of polymethyl methacrylate in a narrow channel well. It is shown that Gpyro models multidimensional conduction within solids well, and that it takes mass that is burned away into consideration. In all 20 tests were done on the International Space Station and their results categorized according to the way the fuel sample was burnt, i.e., one sided or two sided. Oxygen concentration study was done for the experiments. From the results of the tests we observed width effects wherein for the same thickness the wider fuels (2 cm wide) showed a higher spread rate than the narrower fuels (1 cm wide). The 2 cm wide fuels spread faster by 0.045 mm/s on average. Also it was observed that the two sided burns spread faster than the one sided burns by 0.0275 mm/s on average. Agreement with results from the Narrow Channel apparatus was reasonable. For the flame spread in vertical narrow channels, the spread rate monotonically decreased when convection was mixed but not when convection was natural. Flame extinction in a natural convection setting was mostly due to quenching under lunar gravity and perhaps a combination of blow-off and quenching under normal gravity. The vertical velocity values were sufficient to confirm that blow-off was partially responsible for flame extinction under normal gravity. The amount of heat lost to the atmosphere is known, but the cause of heat loss is still not certain. Rendering the right wall of the

	channel adiabatic accelerated the spread rate and prolonged combustion before imminent extinction.
Bibliography Type:	Description: (Last Updated: 06/15/2021)
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